

Raj Patel

Cybersecurity Research | AI Security | Threat Intelligence
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EDUCATION

Georgia Institute of Technology

Master of Science, **Computer Engineering**

Bachelor of Science, **Computer Engineering**

Atlanta, GA

December 2026

December 2024

CERTIFICATIONS

CompTIA Security+, AWS Certified Cloud Practitioner, AWS Certified AI Practitioner

EXPERIENCE

Georgia Tech - Center for Cyber Operations and Unconventional Sensing (COEUS)

May 2026 – Present

Graduate Research Assistant

Atlanta

- Built an agentic AI framework that generates per-sample execution scripts to detonate evasion-aware malware that stays dormant under standard sandbox analysis, raising behavioral coverage on held-out samples from 50% to 83%, and cutting manual triage from 3 hours to 30 minutes per sample.
- Reproduced 3 published binary decomposition research artifacts on a custom software supply chain dataset of 19 Linux applications, evaluating the robustness of reported accuracy on statically linked and stripped binaries.

Fortune 100 Media & Entertainment Company- name withheld per NDA

January 2026 – May 2026

Cyber Threat Intelligence Analyst

Atlanta

- Built a threat intelligence inventory of 14 incidents spanning 3 years of adversary activity, mapping TTPs to MITRE ATT&CK using Recorded Future and OSINT, and briefed findings to leadership.
- Cut research time per incident from 6 hours to 1 by developing and evaluating prompts for Recorded Future AI Insights to surface relevant reporting, refining prompts against an evaluation set of 4 queries.

Pramukh Transport Ltd.

August 2022 – January 2025

Cyber Risk Analyst, Part-time

Kenya

- Conducted a red-team tabletop exercise and threat model across 3 attack objectives, identifying exposed RDP and weak credential management as top risks to transport operations and driving MFA adoption to strengthen access controls.

PROJECTS & RESEARCH

Malware Reverse Engineering

- Reverse engineered 5 malware families in IDA Pro and x64dbg to extract C2 infrastructure, Windows API abuse patterns, and sandbox evasion techniques.
- Analyzed captured traffic in Wireshark to recover C2 protocol structure and exfiltration methods, published as 3 technical writeups for threat intelligence reporting.

ML-Based Network Intrusion Detection System

- Removed hidden backdoor triggers from a trained intrusion detection model using machine unlearning, recovering 98% clean accuracy while dropping attack success rate from 96% to 8%.
- Wrote 3 SOC-ready Suricata signatures for SSH brute-force detection, validated at 90% detection rate against 50 GB of PCAP data.

Secure Cloud Web Application Development

- Enhanced the security of a cloud-native web application by mitigating OWASP Top 10 broken authentication risks through AWS IAM least-privilege RBAC, Cognito authentication, and API Gateway rate limiting.

TECHNICAL SKILLS

- Programming:** Python, C, C++, x86/x64 Assembly, Bash, PowerShell, SQL, KQL
- AI/ML:** Agentic AI, LLM security, model red teaming, RAG, MCP, prompt engineering, PyTorch, scikit-learn, pandas, NumPy
- Security:** IDA Pro, Ghidra, x64dbg, YARA, Wireshark, Suricata, Snort, Splunk, Elastic, Recorded Future, ThreatConnect, DomainTools, Shodan, VirusTotal, Maltego
- Cloud & Infra:** AWS, Docker, Kubernetes, VMware, Linux, Git/GitHub Actions
- Frameworks:** MITRE ATT&CK, MITRE ATLAS, NIST CSF, OWASP Top 10, OWASP LLM Top 10, Cyber Kill Chain, Diamond Model